Bassem Mohammed

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PROFESSIONAL SUMMARY

Artificial Intelligence and Machine Learning graduate and Boston University M.S. Computer Science & AI student, on a co-op placement at VCycene Inc., where I own the data and decision side of a 12-week reinforcement learning study on in-vessel composting machines. I built the sensor pipeline that records the process every 42 seconds, the live decision loop that picks a drying setting every four hours, and the offline evaluation that judges a candidate policy before it reaches a machine. The study replaces two fixed operating modes with one policy learned from sensor data, so the system can hold the compost in the moisture and aeration band that keeps decomposition aerobic. I came to machine learning from mechanical engineering and eight years of operations, so instrumentation and process discipline are familiar ground.

TECHNICAL SKILLS

Languages & Tools: Python · C · Java · SQL · NumPy · Pandas · scikit-learn · Git · pytest · VS Code

Machine Learning: PyTorch · TensorFlow · LightGBM · Gradient Boosting · CNNs · Transfer Learning · Hyperparameter Tuning · Cross-Validation

Reinforcement Learning & Evaluation: Offline RL · Fitted Q Iteration · Discrete IQL · Contextual Bandits · Q-Learning · Propensity Logging · Doubly Robust Off-Policy Evaluation

Bioprocess Instrumentation: BME688 Environmental Sensing · UART and Binary Serial Protocols · Psychrometrics · Thermal and Airflow Control · Time-Series Feature Engineering

Data & Infrastructure: PostgreSQL · Supabase · Row-Level Security · Schema Versioning · CSV Validation · FastAPI

WORK EXPERIENCE

Machine Learning Engineer Intern **Sep 2026 – Dec 2026**
VCycene Inc., Markham, ON

- Own the data and decision side of a 12-week reinforcement learning study on bench composting machines, replacing two fixed drying modes with one policy learned from sensor data.
- Created a verified system and hardware pipeline, then built the offline training and evaluation stack that turns its logs into candidate policies.
- Withdrew two of my own findings after re-checking them against the run output.
- Picked up a colleague's unfinished controller and carried it from seven failing tests to a passing suite inside his original harness.
- Delivered the tooling the team relies on, including bench scripts, the hosted scoring pages, an insights dashboard and the run guide.

Radioactive Engineering Technologist **Aug 2023 – Dec 2025**
London Health Sciences Centre, London, ON

- Analyzed system performance data to resolve recurring faults early and maintained detailed incident logs, habits that carry directly into model monitoring and debugging.

Manufacturing Engineering Co-op **May 2023 – Aug 2023**
ATS Automation, Cambridge, ON

- Analyzed production workflows and diagnosed 5+ recurring faults by root-cause analysis, documenting 20+ process improvement packages with a 10+ member cross-functional team.

Manufacturing Engineering Co-op **Sep 2022 – Dec 2022**
SAF-Holland Canada Limited, Woodstock, ON

- Reverse-engineered and redesigned 3 systems while resolving 10+ recurring faults, producing the process documentation packages that recorded each fix.

Workshop Manager **May 2011 – May 2019**
MAN Truck & Bus, Middle East & Africa Region

- Led 30+ technicians over 8 years, delivering a 20% efficiency gain through data-driven optimization and root-cause frameworks that cut repeat failures by roughly 30%.

PROJECTS

Live Reinforcement Learning Loop on Bench Machines | VCycene Inc.

- Run the live decision loop on the machines. Every four hours each chamber draws one of five drying settings from a guardrail-restricted set, and the harness applies it and records the probability it was chosen with.
- Built the guardrails that keep live exploration safe. A dry score narrows the set to the two gentlest settings, a wet or sour score to the two strongest, and a stale score to the middle three, so a failing chamber is never pushed further.
- Wrote the harness that drives blower duty, paired PTC heating, pad thermostats and mixing cycles over a 16-byte serial protocol, with hardware clamps and a safe state on link loss.

Offline Policy Learning and Evaluation from Logged Data | VCycene Inc.

- Designed the logging contract behind offline evaluation. Every decision records the probability it was chosen with, so a candidate policy can be scored against history without running it on a machine.
- Trained three offline learners on the logged decisions: fitted Q iteration with gradient boosting, discrete IQL in PyTorch, and a linear contextual bandit.
- Judged each candidate by doubly robust off-policy estimates, with cross-fitted reward models and errors clustered by machine and cycle. Under leave-one-machine-out rotation IQL beat both fixed modes on two of three machines, though all learners sat within noise of the baseline on 167 scored decisions.

Streaming Data Pipeline and Training Data Integrity | VCycene Inc.

- Shipped the logging and storage layer for a multi-month study. BME688 readings arrive over UART every 42 seconds and land in PostgreSQL on Supabase, which the training pipeline reads back.
- Parsed a binary serial protocol with frame synchronization and resync, so a byte loss recovers instead of corrupting the record. Caught three sensor faults, and made a stalled USB link reset itself rather than freeze the run.
- Blocked label leakage by construction. A pytest check fails the build if any human-entered column reaches the model inputs, backed by a suite of over 330 tests.

Compost Simulation and Fitted Dynamics | VCycene Inc.

- Prototyped a simulated composter and a tabular Q-learning agent in Python and NumPy before hardware was available, to establish whether an agent could learn in that environment at all.
- Found that the simulator's physics was assumed rather than measured, so its results described the model and not the machine. Documented that plainly and moved the work off the critical path instead of letting it steer the hardware study.

Compost Scoring and Insights Web Application | VCycene Inc.

- Shipped the hosted pages the study runs on: a full compost check with photos and history, a quick daily score, a per-machine insights dashboard and the run guide.
- Charted sensor traces against the decisions that caused them, flagging a late decision or an open lid with a banner so a scorer sees why a reading looks wrong.
- The pages reach Supabase with a restricted key under row-level security that allows insert and select but not update, delete or drop.

EDUCATION

Online M.S. in Computer Science & Artificial Intelligence **Sep 2026 – Present**
Boston University, Boston, MA | Coursework: Foundations of Programming and Systems (C, Java), Mathematical Foundations of Machine Learning

Graduate Certificate in Artificial Intelligence and Machine Learning Co-op **Jan 2026 – Dec 2026**
Fanshawe College, London, ON | GPA 4.18/4.2 | Dean’s Honour Roll

Applied Data Science and Artificial Intelligence **May 2025 – Aug 2025**
MIT Professional Education

B.Eng. in Mechanical Engineering **Sep 2004 – Jul 2010**
Helwan University, Cairo, Egypt

TRAINING AND CERTIFICATIONS

Machine Learning, Stanford and DeepLearning.AI (Dec 2024) · Deep Learning, DeepLearning.AI (Mar 2025) · Mathematics for Machine Learning, Imperial College (Aug 2025) · AI Agents Intensive, Google (Nov 2025)